

2. Research Method

2.1 Introduction

The present study investigates the influence of **Deep Tech-based digital transformation on the quality and efficiency of university education in emerging contexts**. This phenomenon is inherently multidimensional because it simultaneously involves technological innovation, educational processes, organizational transformation, institutional governance, and human experiences. Therefore, selecting an appropriate research method requires evaluating alternative methodological approaches through a systematic decision framework rather than relying on personal preference.

Following the recommendations presented in Session 2, this chapter applies the **Method-Fit Matrix**, a structured evaluation framework based on five methodological criteria:

- Epistemological Fit
- Data Availability
- Feasibility
- Contribution Type
- Venue Fit

This framework enables the comparison of multiple candidate research methods and provides an explicit justification for selecting the methodology that offers the highest scientific coherence.

2.2 Refined Research Question

The research question developed during Session 1 was intentionally broad in order to remain compatible with multiple research paradigms. However, Session 2 emphasizes that a doctoral researcher must progressively refine the question until it becomes method-ready, allowing a direct connection between the research objectives, data collection procedures, and expected scientific contributions.

Accordingly, the refined research question is formulated as follows:

How does Deep Tech-based digital transformation influence educational quality and institutional efficiency in universities operating within emerging contexts, and which organizational, technological, and human factors explain successful digital transformation?

This refined question introduces three important methodological characteristics.

First, it investigates **causal influence** by examining the relationship between digital transformation and institutional outcomes.

Second, it recognizes that educational transformation cannot be understood solely through numerical indicators because organizational culture, leadership, faculty adaptation, and student engagement also shape institutional success.

Third, it explicitly incorporates explanatory elements ("which factors explain"), allowing both quantitative measurement and qualitative interpretation.

Consequently, the research question naturally supports an integrated methodological approach.

2.3 Research Purpose

The purpose of this study is to explain how Deep Tech-enabled digital transformation contributes to improving educational quality and institutional efficiency within universities located in emerging economies by integrating quantitative institutional performance indicators with qualitative evidence derived from the experiences of academic stakeholders.

Rather than merely evaluating technology adoption, the study seeks to understand the mechanisms through which Deep Technologies create organizational and educational value.

2.4 General Objective

To analyze the influence of Deep Tech-based digital transformation on the quality and operational efficiency of university education in emerging contexts.

2.5 Specific Objectives

The research pursues the following objectives:

1. To evaluate the relationship between Deep Technology adoption and educational quality.
2. To determine the influence of digital transformation on institutional efficiency.
3. To identify organizational and human factors facilitating successful digital transformation.
4. To integrate quantitative institutional indicators with qualitative stakeholder experiences in order to explain digital transformation outcomes.

5. To formulate evidence-based recommendations for sustainable digital transformation strategies in higher education institutions.

2.6 Methodological Alignment

One of the central principles emphasized during Session 2 is that **method follows paradigm**. Consequently, methodological decisions should never be made independently from the epistemological assumptions established during paradigm selection.

The methodological alignment of the present research is summarized below.

Component	Alignment
Research Topic	Deep Tech-Based Digital Transformation
Research Paradigm	Mixed Methods
Research Design	Explanatory Sequential Design
Research Question	Influence + Explanation
Evidence	Quantitative + Qualitative
Expected Contribution	Explanatory Framework for Digital Transformation
Target Publications	Computers & Education; IEEE Transactions on Education; Education and Information Technologies

The proposed alignment demonstrates complete consistency between the research problem, the selected paradigm, and the intended scientific contribution. Unlike studies that merely compare technological performance indicators, this investigation seeks to explain both **whether** Deep Tech improves university performance and **why** these improvements occur under specific organizational conditions.

Such alignment represents the methodological foundation required for rigorous doctoral research and reflects the principles established throughout Session 2.

2.7 Candidate Research Methods

Following the Method-Fit Matrix proposed during Session 2, three candidate research methods were identified as theoretically viable alternatives for investigating the research problem.

These methods are:

- Candidate Method 1: Quasi-Experimental Research
- Candidate Method 2: Multiple Case Study
- Candidate Method 3: Explanatory Sequential Mixed Methods

The following sections evaluate each alternative using the five methodological criteria introduced during the doctoral seminar before selecting the most appropriate research strategy.

2.7 Evaluation of Candidate Research Methods

Following the methodological framework proposed in Session 2, three research methods were identified as viable alternatives for investigating the influence of Deep Tech-based digital transformation on the quality and efficiency of university education in emerging contexts. The selection of these methods was guided by the principle that research methods must be aligned with the chosen paradigm, research objectives, expected scientific contribution, and practical feasibility. Rather than selecting a method based solely on familiarity or disciplinary tradition, this study systematically evaluates each alternative according to its capacity to answer the refined research question.

2.7.1 Candidate Method 1: Quasi-Experimental Research

Quasi-experimental research represents one of the most frequently used methodologies in Artificial Intelligence, educational technology, and digital innovation studies because it allows researchers to evaluate the effects of interventions under real-world conditions when random assignment is impractical or ethically impossible. In higher education, universities cannot usually be randomly assigned to adopt or reject digital transformation initiatives. Consequently, quasi-experimental designs provide a practical alternative for assessing institutional change.

Within the context of this research, a quasi-experimental design could compare universities that have implemented Deep Technology solutions with institutions that continue using conventional digital systems. Institutional indicators such as

student academic performance, graduation rates, learning analytics, administrative processing time, digital service utilization, and operational efficiency could be collected before and after implementation or compared across institutions with different levels of digital maturity.

The principal advantage of this method is its capacity to generate objective and statistically measurable evidence regarding the influence of Deep Technologies. Advanced statistical techniques, including multiple regression analysis, structural equation modeling, multilevel analysis, or propensity score matching, could be employed to estimate the magnitude of digital transformation effects while controlling for institutional differences.

Despite these strengths, the quasi-experimental approach presents important limitations. Educational transformation is not solely a technological process but also an organizational and cultural phenomenon. Statistical analyses may reveal that institutional performance has improved after implementing Artificial Intelligence or learning analytics systems, yet they cannot adequately explain why some universities achieve better outcomes than others despite adopting similar technologies. Variables such as institutional leadership, faculty readiness, organizational culture, digital competencies, and stakeholder engagement remain largely invisible within purely quantitative analyses.

Consequently, although quasi-experimental research offers robust empirical evidence regarding institutional performance, it provides only a partial explanation of the mechanisms through which Deep Technologies influence educational quality and institutional efficiency.

Expected Scientific Contribution

This method would primarily generate:

- Statistical evidence regarding the effects of Deep Technologies on university performance.
- Quantitative estimates of educational quality improvement.
- Objective indicators of institutional efficiency.
- Empirical relationships between technology adoption and organizational outcomes.

2.7.2 Candidate Method 2: Multiple Case Study

The multiple case study approach focuses on understanding complex organizational phenomena within their real-world environments. According to Yin (2018), case study research is particularly appropriate for investigating "how" and "why" questions where contextual factors cannot be separated from the phenomenon under investigation.

For the present study, several universities implementing Deep Technology initiatives could be selected as individual cases. Data collection would involve multiple sources of evidence, including institutional strategic plans, digital transformation policies, interviews with university administrators, faculty members, information technology specialists, and students, together with classroom observations and institutional performance reports.

The principal strength of the case study method lies in its capacity to capture organizational complexity. Universities differ considerably in terms of governance structures, technological infrastructure, institutional culture, financial capacity, and leadership strategies. A case study enables the researcher to understand how these contextual characteristics shape digital transformation processes and influence educational outcomes.

Furthermore, multiple case studies facilitate cross-case comparison. Similarities and differences among institutions can be analyzed to identify critical success factors and barriers affecting the implementation of Deep Technologies in higher education.

However, the method also presents limitations. Although case studies generate rich contextual knowledge, they provide limited quantitative evidence regarding the magnitude of digital transformation effects. The findings are context-dependent and cannot easily support statistical generalization across a broader population of universities. Consequently, while the method explains organizational processes effectively, it is less suitable for demonstrating measurable institutional improvements.

Expected Scientific Contribution

A multiple case study would primarily contribute:

- Comprehensive understanding of organizational digital transformation.
- Identification of institutional best practices.
- Development of explanatory organizational frameworks.

- Contextual knowledge supporting future comparative research.

2.7.3 Candidate Method 3: Explanatory Sequential Mixed Methods

The explanatory sequential mixed methods design integrates quantitative and qualitative research within a single coherent methodological framework. According to Creswell and Creswell (2018), this design begins with quantitative data collection and analysis, followed by a qualitative phase specifically designed to explain the statistical findings.

This methodological strategy is particularly appropriate for investigating digital transformation because universities simultaneously represent technological systems, educational organizations, and social communities. Consequently, understanding digital transformation requires both objective institutional indicators and stakeholder perspectives.

During the quantitative phase, institutional performance indicators would be collected from participating universities. These indicators may include student academic achievement, graduation rates, student retention, learning analytics, administrative productivity, operational costs, technology adoption indices, and institutional efficiency metrics. Statistical analyses would determine whether significant relationships exist between Deep Technology implementation and educational performance.

The second phase would involve qualitative data collection through semi-structured interviews with university administrators, faculty members, information technology managers, instructional designers, and students. These interviews would seek to explain why quantitative differences emerged by exploring organizational culture, leadership practices, faculty adaptation, digital competencies, institutional innovation, and stakeholder experiences.

Finally, both datasets would be integrated during the interpretation stage. Rather than treating quantitative and qualitative findings as independent studies, the explanatory sequential design combines them to produce a comprehensive explanation of digital transformation. Statistical evidence identifies **whether** Deep Technologies improve educational quality and institutional efficiency, while qualitative evidence explains **how** and **why** these improvements occur.

This integration represents the principal strength of the explanatory sequential design because it produces knowledge that neither quantitative nor qualitative methods could generate independently.

Expected Scientific Contribution

The explanatory sequential mixed methods approach is expected to generate:

- Integrated quantitative and qualitative evidence.
- A comprehensive explanation of Deep Technology implementation.
- An explanatory framework connecting technological innovation, organizational transformation, and educational quality.
- Practical recommendations supporting sustainable digital transformation strategies for higher education institutions in emerging contexts.

2.7.4 Comparative Assessment of the Candidate Methods

The three candidate methods provide different perspectives on the research problem. Quasi-experimental research offers rigorous statistical evidence regarding institutional performance but provides limited understanding of organizational dynamics. Multiple case studies produce rich contextual explanations but lack sufficient quantitative evidence to demonstrate institutional impact across universities. In contrast, the explanatory sequential mixed methods design combines the strengths of both approaches by integrating objective performance indicators with in-depth stakeholder experiences.

Given the multidimensional nature of Deep Tech-based digital transformation, the explanatory sequential mixed methods design appears to offer the highest level of methodological coherence. Nevertheless, a systematic evaluation using the **Method-Fit Matrix** is required before making the final methodological decision. The following section applies this evaluation framework by assessing each candidate method according to five criteria: Epistemological Fit, Data Availability, Feasibility, Contribution Type, and Venue Fit.